

MAX MRI RADIOLOGY, A Medical Group Inc.

PATIENT NAME: **HANDROP, RONALD** DOB: **12/07/1954**
REFERRING PHYSICIAN: **BLUMENFELD, ANDREW** DATE OF EXAM: **11/20/2023 02:20:03 PM**
STUDY DESCRIPTION: **MR BRAIN DTI/TBI** MRI MACHINE: **Philips Evolution 3T**

MRI OF THE BRAIN WITHOUT CONTRAST:

CLINICAL INDICATION: Motor vehicle accident 10/18/2023. Concussion.

TECHNIQUE: T1, T2, FLAIR, diffusion, and GRE images were obtained. Neuroquantitative Analysis was performed. Susceptibility weighted images were obtained. Diffusion tensor images were obtained.

FINDINGS:

SUPRATENTORIAL STRUCTURES: There are multiple patchy and confluent foci of white matter disease.

There are patchy confluent foci of T2/FLAIR prolongation within the supratentorial white matter. Given the patient's indication, the differential diagnosis includes trauma/traumatic brain injury, as well as other etiologies such as vasculitis and migraine headache which should be ruled out clinically. There is no mass, mass effect, or hemorrhage.

POSTERIOR FOSSA: There are findings suspicious for chronic left cerebellar infarction.

The brainstem is normal in signal intensity. The cerebellum appears within normal limits. The internal auditory canals appear within normal limits. The seventh and eighth cranial nerves are normal and there is no evidence for cerebellopontine angle mass.

VENTRICULAR SYSTEM: The ventricles are normal in size and shape. There is no evidence for hydrocephalus and there is no evidence for transependymal flow of CSF.

SKULL BASE AND OSSEOUS STRUCTURES: There is a right lens replacement. There is paranasal sinus mucosal thickening, most prominent in the right maxillary sinus. Clear temporal bone structures. The sella structures appear within normal limits. There is no evidence for abnormal mass or fluid collection associated with these structures.

VASCULAR STRUCTURES: There is normal signal void within the major vessels of the circle of Willis. The superior sagittal sinus appears unremarkable on this examination.

SWI: The susceptibility weighted images demonstrate no areas of signal dropout to suggest residual blood products.

DIFFUSION TENSOR IMAGING: Whole brain analysis of the DTI sequences demonstrates no areas of reduced fractional anisotropy. It is important to note that negative DTI sequences do not rule out the clinical diagnosis of mild traumatic brain injury.

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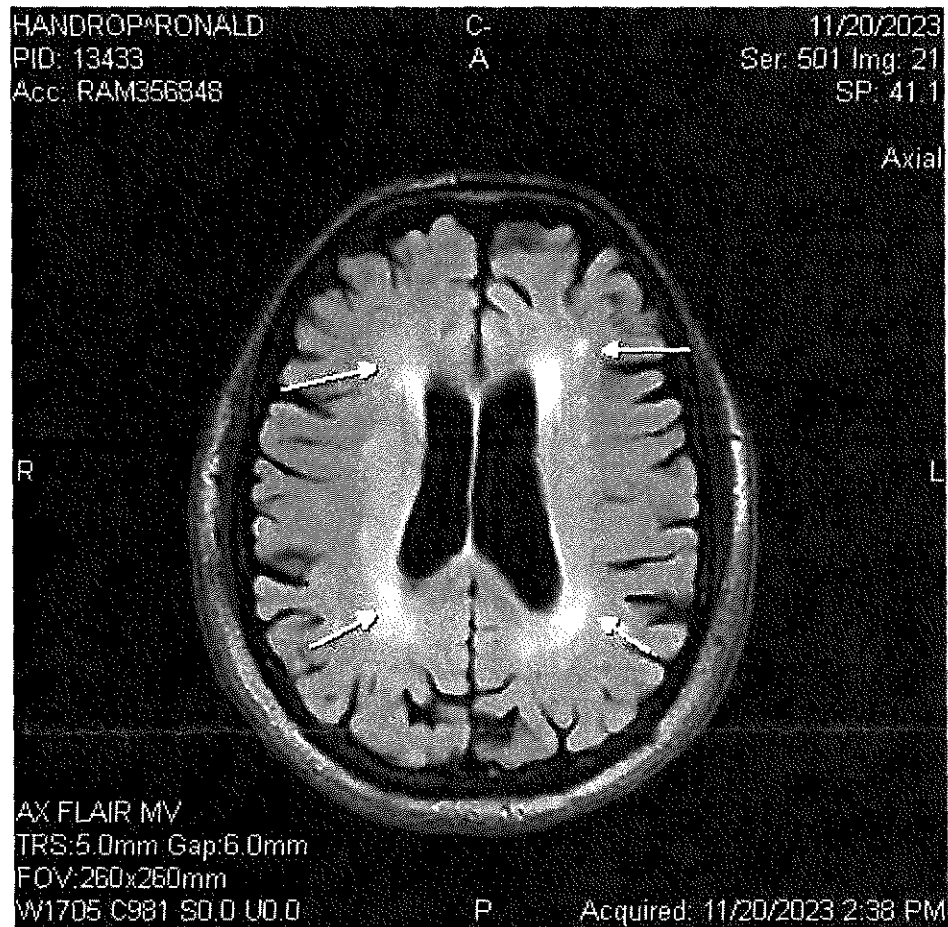
NEUROQUANT ANALYSIS: Analysis of brain volumes was performed. The left hippocampus measures 2.90 cubic centimeters and the right hippocampus measures 2.98 cubic centimeters. These values are within the normative ranges for the patient's age. There is no evidence for atrophy of the medial temporal lobes.

IMPRESSION:

1. There are multiple patchy and confluent foci of white matter disease.
2. There are patchy confluent foci of T2/FLAIR prolongation within the supratentorial white matter. Given the patient's indication, the differential diagnosis includes trauma/traumatic brain injury, as well as other etiologies such as vasculitis and migraine headache which should be ruled out clinically. There is no mass, mass effect, or hemorrhage. *See **Figure 1** Axial T2 FLAIR image #21. Arrows show the patchy confluent foci of T2/FLAIR prolongation within the supratentorial white matter.
3. There are findings suspicious for chronic left cerebellar infarction.

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FIGURE 1:



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Thank you for the consultation.

Sincerely,



Matthew D. Hermann, M.D.

Neuro/Pain Specialist, ExpertRadiology

Board Certified, Pain Management, American Board Of Radiology

Board Certified, Diagnostic Radiology, American Board Of Radiology

Fellowship Trained, Stanford Division of Pain Medicine

Avery J. Knapp Jr., M.D., Medical Director, ExpertRadiology

Electronically Signed: Electronically Signed by HERMANN, MATTHEW MD DR. at 11/22/2023 10:19:36 AM
Transcribed/ D.J.

NeuroQuant® TBA

Triage Brain Atrophy Report: Single Timepoint

Patient Information

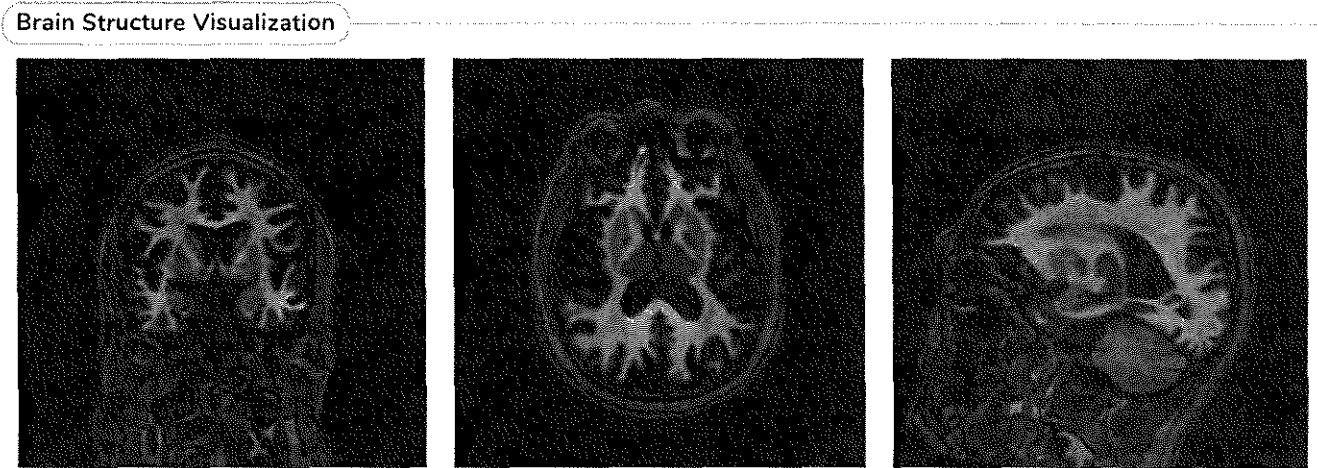
Patient: HANDROP, RONALD
Referring Physician: BLUMENFELD, ANDREW
Age: 68 Sex: M Patient ID: 13433

Report Information

Scan Date: 2023-11-20

Site Information

MAX MRI RADIOLOGY
Anna Rodriguez



Brain Structure Volumes

Brain Volumes	Volume (cm³)	Normative Percentile
Whole Brain	1119	45
Cortical Gray Matter	445	41
Cerebral White Matter	468	46

Regional Brain Structures

Regional Brain Structures	Normative Percentiles		
	Left	Right	Total
Ventricles	97	84	94
Cerebellum	67	67	67
Brainstem	-	-	55
Deep Gray Nuclei			
- Putamen	55	60	58
- Thalamus	86	78	83
- Caudate	89	80	86
Cingulate	27	6	8
- Anterior Cingulate Gyrus	43	7	11
- Posterior Cingulate Gyrus	30	29	25

Color Code Key:

A structure is below the 5th percentile OR a ventricle is above the 95th percentile.

Regional Lobe Analysis

Regional Lobe Analysis	Normative Percentiles		
	Left	Right	Total
Temporal Lobe	60	59	60
- Hippocampus	37	36	36
- Entorhinal Cortex	99	99	99
- Parahippocampal Gyrus	94	83	92
- Amygdala	58	68	64
- Fusiform Gyrus	74	62	71
- Superior Temporal Gyrus	71	86	81
- Middle Temporal Gyrus	46	14	27
- Inferior Temporal Gyrus	5	2	1
- Temporal Pole	87	99	99
Parietal Lobe	49	33	39
- Primary Sensory Cortex	58	69	66
- Superior Parietal Lobule	72	66	70
- Supramarginal Gyrus	32	12	19
Frontal Lobe	37	47	42
- Primary Motor Cortex	4	16	7
- Superior Frontal Gyrus	21	81	54
- Middle Frontal Gyrus	86	74	82
- Inferior Frontal Gyrus	27	1	6
- Lateral Orbitofrontal Gyrus	11	17	12
- Medial Orbitofrontal Gyrus	92	86	92
Occipital Lobe	72	52	62
- Medial Occipital	73	62	68
- Lateral Occipital	63	40	51